

Introduction to R

<https://tinyurl.com/hbc-r-flipped>



Harvard Chan Bioinformatics Core

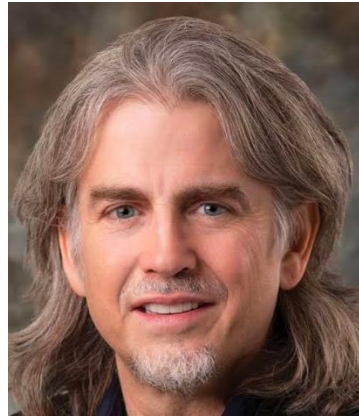


Introductions!





Shannan Ho Sui
Director



John Quackenbush
Faculty Advisor



James Billingsley



Emma Berdan



Elizabeth
Partan



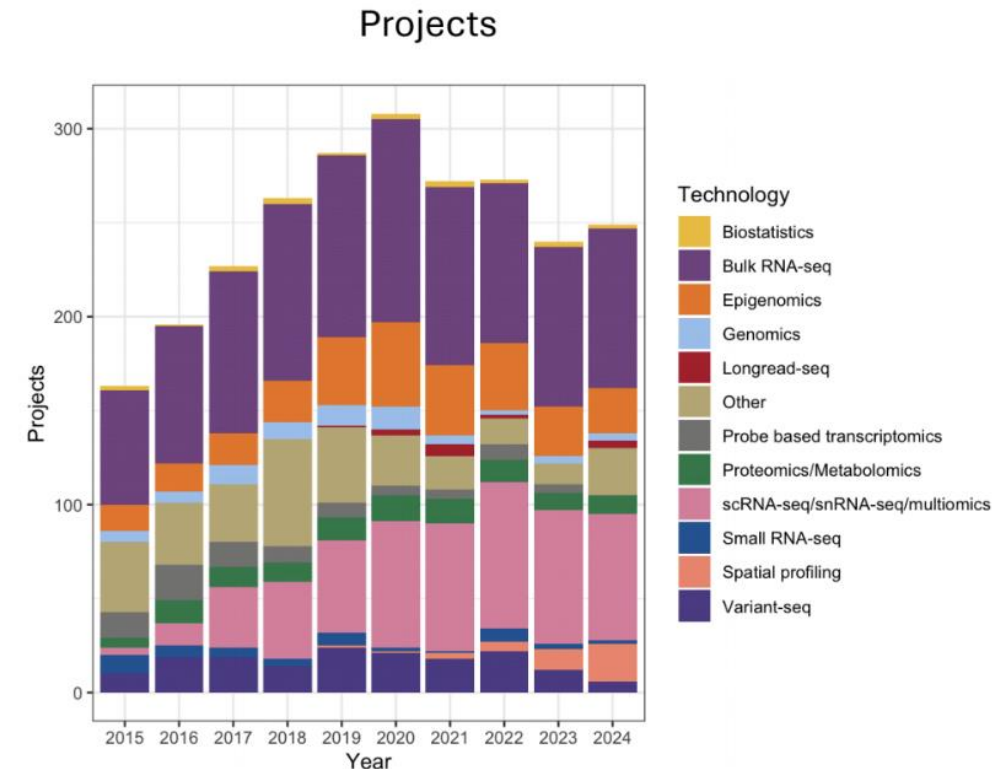
Will Gammerdinger



Noor Sohail

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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NIEHS



HARVARD
CATALYST

THE HARVARD CLINICAL
AND TRANSLATIONAL
SCIENCE CENTER



HARVARD
MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Python

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Current Topics in Bioinformatics

<https://hsph.harvard.edu/research/bioinformatics/training/>

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❖ Basic Data Skills

- ❖ Shell
- ❖ R
- ❖ Python

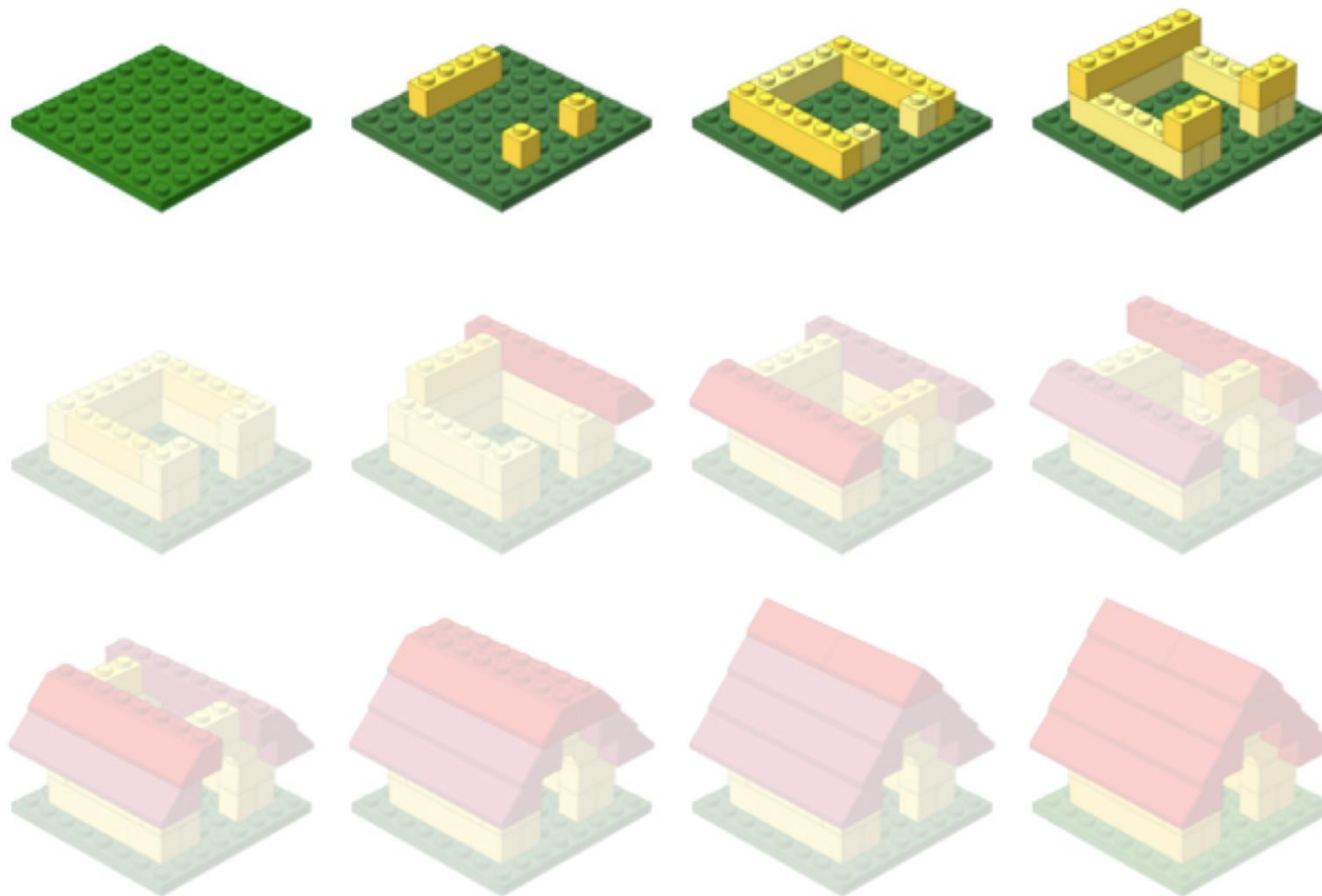
❖ Advanced Topics: Analysis of high-throughput sequencing data

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- ❖ Differential Gene Expression
- ❖ scRNA-seq
- ❖ Variant Calling

❖ Current Topics in Bioinformatics

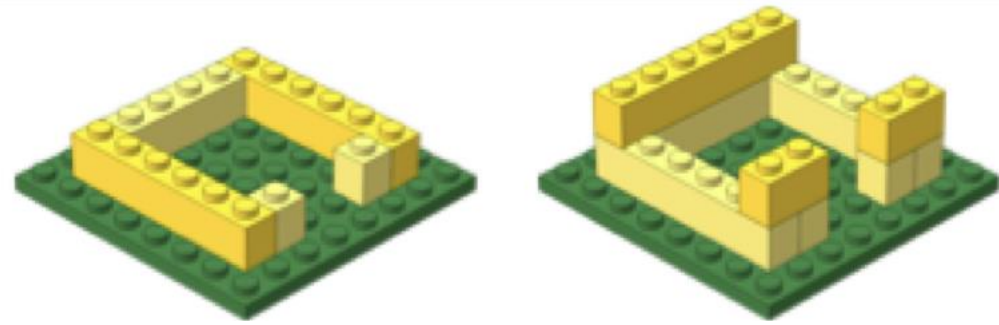


Workshop scope



Learning R

Workshop Scope



- ❖ Comfortably use RStudio (a graphical interface for R)
- ❖ Fluently interact with R using RStudio
- ❖ Become familiar with R syntax
- ❖ Understand data structures in R
- ❖ Inspect and manipulate data structures
- ❖ Install packages and use functions in R

CRAN



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[ABCOptim](#)
[ABCp2](#)
[abcrf](#)

Available CRAN Packages By Name
[A](#)[B](#)[C](#)[D](#)[E](#)[F](#)[G](#)[H](#)[I](#)[J](#)[K](#)[L](#)[M](#)[N](#)[O](#)[P](#)[Q](#)[R](#)[S](#)[T](#)[U](#)[V](#)[W](#)[X](#)[Y](#)[Z](#)

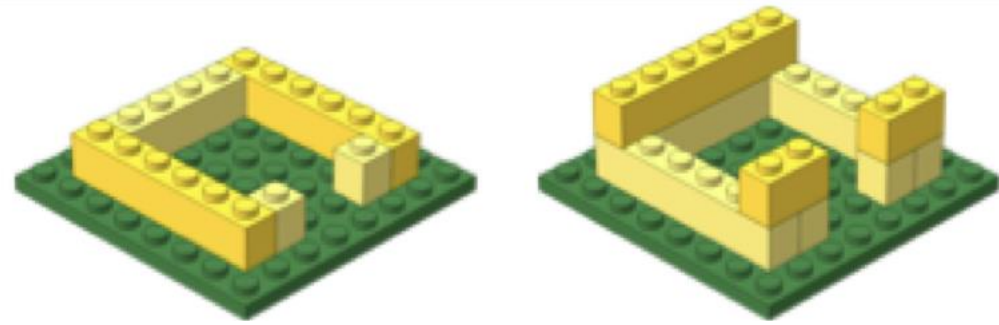
Accurate, Adaptable, and Accessible Error Metrics for Predictive Models
Access to Abbyy Optical Character Recognition (OCR) API
Tools for Approximate Bayesian Computation (ABC)
Computed ABC Analysis
Data Only: Tools for Approximate Bayesian Computation (ABC)
ABCDE_FBA: A-Biologist-Can-Do-Everything of Flux Balance Analysis with this package
Implementation of Artificial Bee Colony (ABC) Optimization
Approximate Bayesian Computational Model for Estimating P2
Approximate Bayesian Computation via Random Forests

- ❖ Comprehensive R Archive Network
- ❖ The main repository for R packages
- ❖ Easy to install

<https://cran.r-project.org/>

- ❖ An alternative package repository; “..provides tools for the analysis and comprehension of *high-throughput genomic data*.”
- ❖ Includes (but is not limited to) tools for:
- ❖ Performing statistical analysis
- ❖ Accessing public datasets
- ❖ Open source and open development
- ❖ Free

Workshop Scope



- ❖ Comfortably use RStudio (a graphical interface for R)
- ❖ Fluently interact with R using RStudio
- ❖ Become familiar with R syntax
- ❖ Understand data structures in R
- ❖ Inspect and manipulate data structures
- ❖ Install packages and use functions in R
- ❖ Visualize data using ggplot2
- ❖ Utilize pipes, tibbles and functions from the Tidyverse package suite

Logistics



Course schedule

[Introduction to R](#) [Schedule](#) [HBC](#) [GitHub](#) [Contact us](#) 

Schedule

 Code

Workshop Schedule

Day 1

Time	Topic	Instructor
10:00 - 10:30	Workshop Introduction	Will
10:30 - 11:45	Introduction to R and RStudio	Noor
11:45 - 12:00	Overview of self-learning materials and homework submission	Will

Before the next class:

<https://tinyurl.com/hbc-r-flipped>

Course materials

❖ We continuously update our materials to reflect changes in the field/software

[Introduction to R](#) [Schedule](#) [HBC](#) [GitHub](#) [Contact us](#)

Day 1:
[Introduction to R and RStudio](#)

Day 1 Self-learning: >

Day 2 >

Day 2 Self-learning >

Day 3 >

Day 3 Self-learning >

Day 4 >

Introduction to R and RStudio

AUTHOR
Mary Piper, Meeta Mistry

PUBLISHED
September 8, 2017

Approximate time: 45 minutes

Learning Objectives

- Describe what R and RStudio are.
- Interact with R using RStudio.
- Familiarize various components of RStudio.
- Employ variables in R.

What is R?

The common misconception is that R is a programming language but in fact it is much more than that. Think of R as an environment for statistical computing and graphics, which brings together a number of features to provide powerful functionality.

</> Code

On this page

[Learning Objectives](#)

[What is R?](#)

[Why use R?](#)

[What is RStudio?](#)

[Creating a new project directory in RStudio](#)

[RStudio Interface](#)

[Organizing your working directory & setting up](#)

[Interacting with R](#)

[The R syntax](#)

[Assignment operator](#)

[Variables](#)

[Interacting with data in R](#)

[Best practices](#)

<https://tinyurl.com/hbc-r-flipped>

Single Screen & 3 Windows

The image is a collage of three screenshots illustrating a single-screen workflow for bioinformatics data analysis.

Top Left Screenshot: A Zoom meeting window showing participants Noor Sohal and Will Gammerding. Overlaid on the Zoom window is a JupyterLab interface. The JupyterLab interface shows a file browser on the left with folders like 'data', 'figures', 'results', and 'data.zip'. The main code editor shows Python code for reading a CSV file, getting row names, and indexing a DataFrame.

Bottom Right Screenshot: An R console window showing R version 4.5.3, copyright information, and a license notice.

Top Right Screenshot: A web browser displaying a page titled 'omics'. The page has a sidebar with 'Learning objectives' such as 'Overview of lesson', 'High-throughput sequencing overview', 'Sequencing-based technologies', 'Imaging-based technologies', 'Summary of comparison', 'Experimental design considerations', and 'Environmental design considerations'.

*Our
Recommendation*

Single Screen & 3 Windows

Zoom

Meeting: Will Gammerdinger's screen

Participants: Noor Sohal, Will Gammerdinger (host)

Zoom Meeting

You to Everyone 8:38 AM

I have a question about how to make figures in Python. How do you make multiple plots at once?

On this page

Learning objectives

- Overview of lesson
- High-throughput sequencing overview
- Sequencing-based technologies
- Imaging-based technologies
- Summary of comparison
- Experimental design considerations

transcriptomics technologies. Focusing on contrasting forms to show how they differ in resolution, you will be informed on which technology is best suited

MERFISH, seqFISH

roughput sequencing

Environment is empty

Our Recommendation

1:1 (Top Level) Posit AI: No completions available. R Script

Console Terminal Background Jobs

R 4.5.3 · ~/Harvard University Dropbox/HBC Team Folder (1)/Teaching/Training_programs/Wc

R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"

Copyright (C) 2026 The R Foundation for Statistical Computing

Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY. You are welcome to redistribute it under certain conditions. Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

Files Plots Packages Help Viewer Presentation

Folder File Delete Rename

g > Training_programs > Workshops > new_workshop_data > spatial_transcriptomics

Name	Size	Modified
..		
data		
figures		
README.txt	1.9 KB	Apr 29, 2026, 8:39 PM
results		
scripts		
spatial_transcriptomics.Rproj	205 B	Apr 30, 2026, 1:12 PM

Single Screen & 3 Windows

Web Browser

Our Recommendation

Single Screen & 3 Windows

The image is a collage of three screenshots illustrating a single-screen workflow. The top-left screenshot shows a Zoom meeting window with participants Noor Sohal and Will Gammerdinger, overlaid on a JupyterLab interface displaying a Python script for data analysis. The top-right screenshot shows a Zoom chat window with a question about Python plotting, overlaid on a web page titled 'omics' with a sidebar of learning objectives. The bottom screenshot shows the R Studio interface, including the Files, Plots, Packages, Help, Viewer, and Presentation panes, with a large 'R Studio' watermark.

Our Recommendation

Single Screen & 3 Windows

The image displays three overlapping software windows. The top-left window, labeled 'Zoom' in blue, shows a video call in progress with participants Noor Sohal and Will Gammerdinger. The top-right window, labeled 'Web Browser' in green, shows a webpage for 'HBC' with a sidebar menu and a main content area. The bottom window, labeled 'R Studio' in purple, shows the R development environment with a console window displaying R version 4.5.3 and a file explorer on the right.

Zoom

Web Browser

R Studio

Our Recommendation

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all of your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete
- ❖ Homework load is heavier in the beginning of this workshop series and tapers off

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

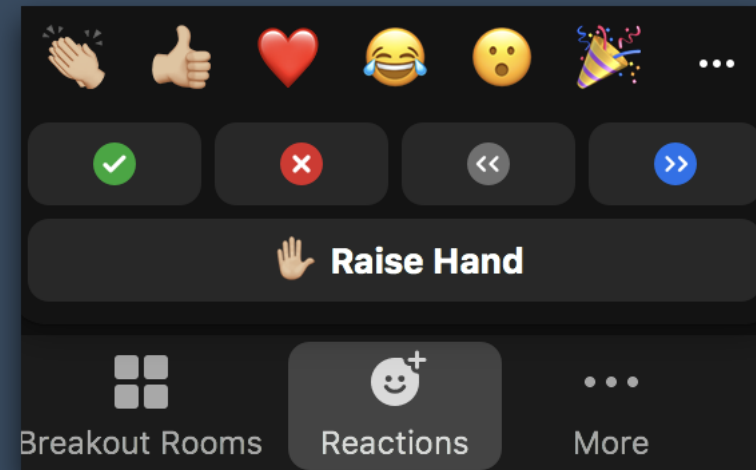
Odds & Ends

❖ Quit/minimize all applications that are not required for class

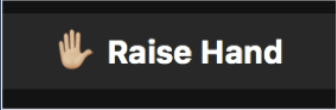
❖ Are you all set?

❖  = "agree", "I'm all set"

❖  = "disagree", "I need help"



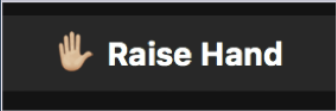
Odds & Ends

- ❖ Questions for the presenter?
 - ❖ Post the question in the Chat window OR
 - ❖  when the presenter asks for questions
 - ❖ Let the Moderator know

Odds & Ends

❖ Questions for the presenter?

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- ❖ Let the Moderator know

❖ Technical difficulties with software?

- ❖ Start a private chat with the Moderator with a description of the problem

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu